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SURGICAL GI DISORDERS IN CHILDREN

Idiopathic Hypertrophic Pyloric Stenosis (IHPS)

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Pathology :

Hypertrophy of the circular muscles in pylorus → Obstruction (Stenosis).

Associations :

- m/c in 1st born male child.
- erythromycin usage.

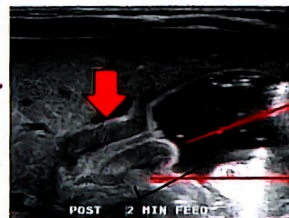
Clinical features :

Onset >3 weeks after birth.

Clinical features	Characteristics
Vomiting	• Non bilious, projectile.
Epigastric swelling	• Usually following episodes of vomiting. • Olive-shaped.
Visible gastric peristalsis	• ↑↑ peristaltic activity to overcome obstruction. • Direction : Left → right. • Seen after feeding.
Dehydration & electrolyte imbalances	D/t chronic vomiting : milk/feeds, H^+ ions, Cl^- ions, K^+ ions ↓ Hypokalemic hypochloremic metabolic alkalosis ↓ If no treatment Na^+ ions lost through vomitus ↓ Kidney reabsorbs Na^+ & secretes K^+ ↓ Depletion of K^+, secretion of H^+ into urine instead ↓ Paradoxical aciduria

Investigations :

1. Ultrasonogram (USG) : IOC.



Pyloric muscle wall thickness ≥4 mm

Pyloric channel length ≥16 mm

IHPD diagnosis based on USG

2. Barium contrast X-ray :

- Not commonly used.
- Narrowing of pyloric channel visualized as "String sign".



String sign

management :

Step 1 : Correction of dehydration.

- Iv fluids used : (1/2 normal saline + K^+) + (5% dextrose).
- Corrects : Hypokalemia, hypoglycaemia, fluid loss.

Step 2 : Surgical treatment → **Ramstedt's pyloromyotomy.**

Duodenal Atresia

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m/c type of intestinal atresia.

Pathology : Failure of recanalization of duodenal lumen.

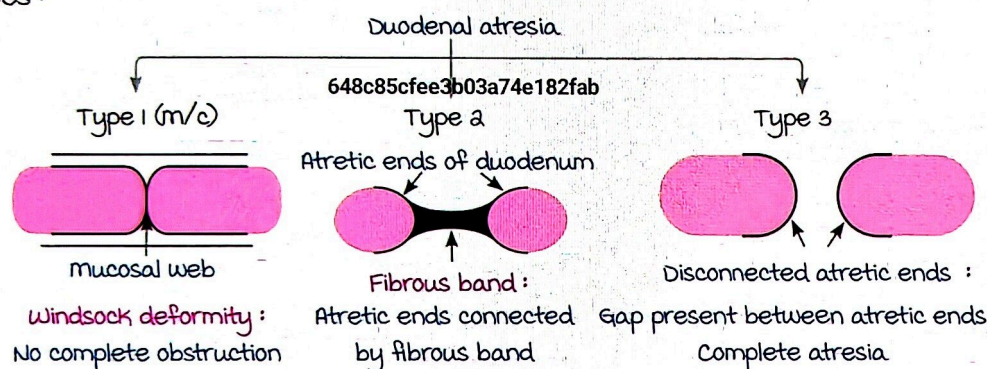
Associations : Down's syndrome.

- Seen in approximately 30% babies with Down's syndrome.

Clinical features :

- Polyhydramnios : Antenatal difficulty in swallowing of amniotic fluid.
- Bilious vomiting : Soon after birth.

Types :



Investigations :

- X-ray :
 - **Double bubble sign** : Also seen in annular pancreas (Obstruction d/t part of pancreas surrounding duodenum).
- USG/CT :
 - **IOC** : Atresia distinctly seen.

Treatment :

- Duodeno-duodenostomy : Atretic ends of duodenum connected.
 - **Preferred method** (↓ risk of post surgery complications like anastomotic leakage).
- Duodeno-jejunostomy : Atretic ends of the duodenum connected to jejunum.



Double bubble sign

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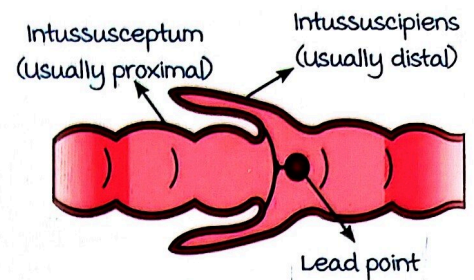
Intussusception

m/c cause of intestinal obstruction in 5 months to 3 years.

m/c cause of abdominal emergency in children
<2 years.

Pathology :

- Telescoping of one part of intestine into another → Obstruction.
- m/c site : Ileocolic junction.



Pathogenesis :

Complimentary feeding/viral infection/vaccination

Lead point mass formed

- <2 years : Swollen Peyer's patches.
- >2 years : Polyp/lipoma.

Telescoping of intestinal mesentery over it

Obstruction to venous return

Edema of bowel + bleeding of mucosa

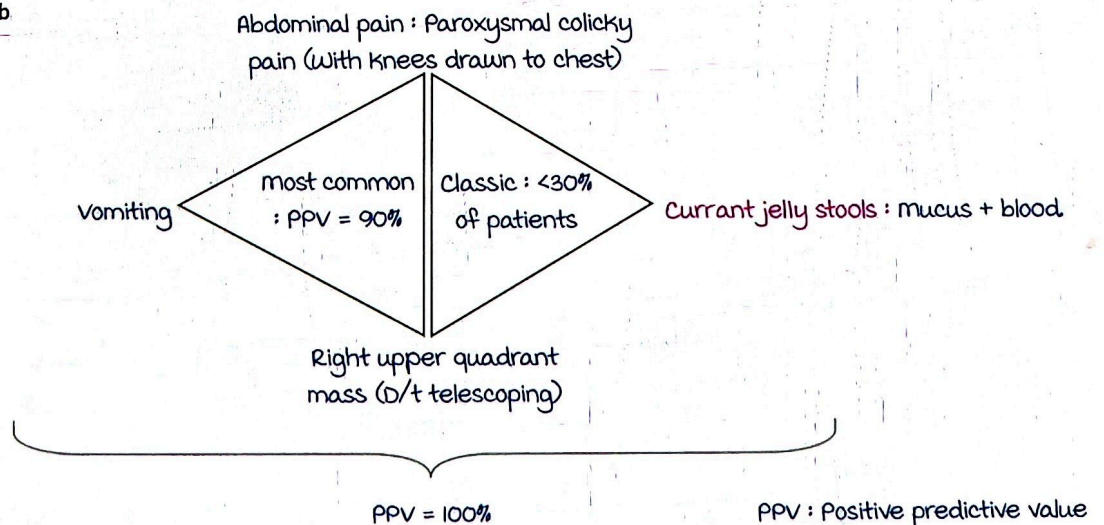
If untreated

Necrosis and/or perforation of bowel

Clinical features :

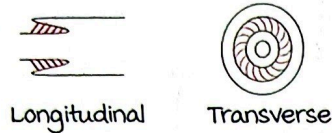
- Age of onset : <2 years (usually around 6 months d/t complimentary foods).
- Seen in previously well fed child.
- Triads seen :

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Investigations :• **USG : (IOC)**

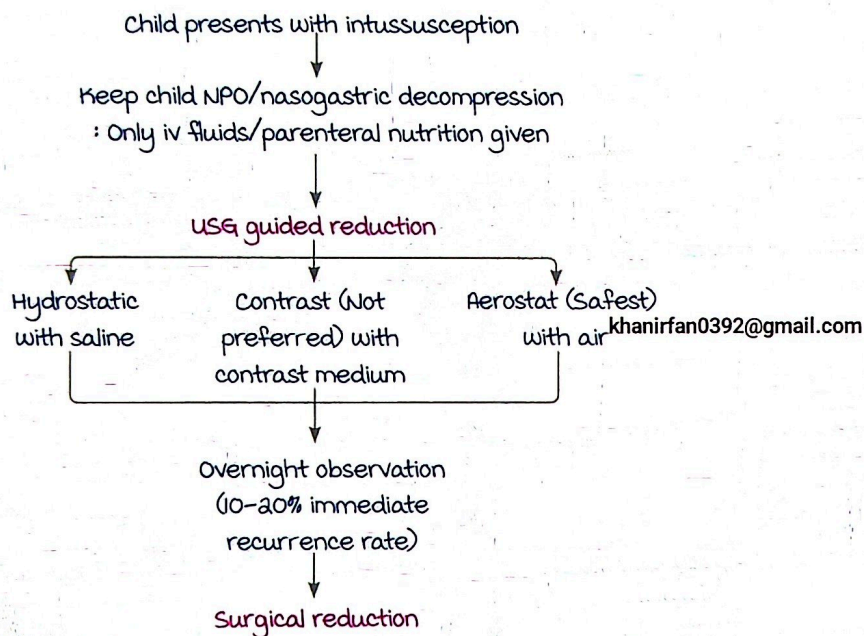
- Longitudinal image : Telescoping seen.
- Transverse image : **Doughnut sign** d/t intussusceptum seen inside intussuscepiens.



- Colour doppler : Done to assess vascularity of affected tissue.
- Barium contrast enema (rarely done) : **Claw sign** seen at site of intussusception.

Treatment :

- Early recognition of intussusception is important.

**Hirschsprung Disease**

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Pathology :

Failure of migration of neurons from neural crest



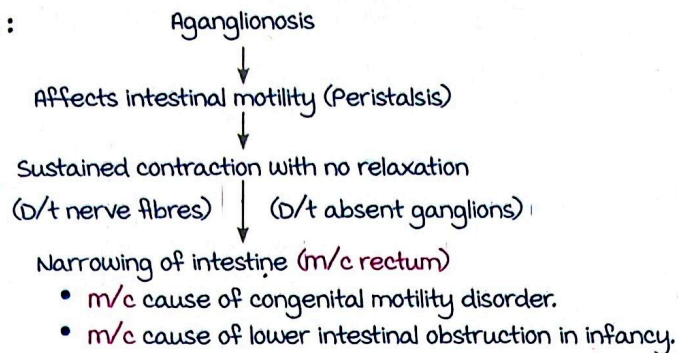
Aganglionosis in submucosal & myenteric plexus.

- m/c site affected : **Rectum > sigmoid.**

Associations :

m/c gastric association : **Down's syndrome.**

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Consequences :**Clinical features :**

1. Delayed passage of meconium : >48 hours.
 2. Abdominal distension.
 3. On per rectal examination
 - Anal sphincter : Normal
 - Lumen : Narrow
- On removal of finger → Sudden expulsion of meconium d/t transient dilatation

Investigations :

1. Radiology :

a. Plain X-ray : ↓/absent air in the pelvis.

b. Barium contrast enema : (Diagnostic) :
Segments may not always be visualized.

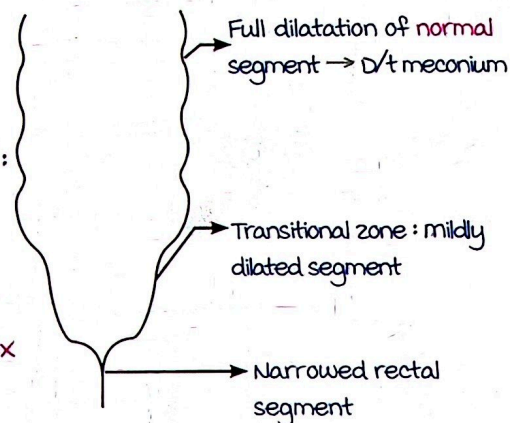
2. Rectal manometry :

- Absence of *recto-anal inhibitory reflex* (RAIR) : No decrease in anal tone on rectal dilation.

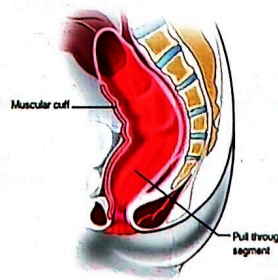
3. Rectal biopsy : *IOC*.

- Ganglions : Absent.
- Nerve fibres :

648c85cfee3b03a74e182fab Hypertrophied (D/t lack of inhibitory effects of ganglions).
- Stained with *acetylcholinesterase*.

**Treatment :**

Early presentation
↓
Pull-through procedure : Removal of affected segment
↓
Unaffected proximal segment pulled through & attached to anus



Pull through procedure

Delayed presentation :
Constipation in a older child
↓
Colostomy : Relieves obstruction
↓
Pull-through procedure